

Robotics and Computer Vision

 Spring 2026

 **Instructors:** Prof. Hamid D. Taghirad & Mohammad Javad Ahmadi

Course Description

This course reinforces and consolidates its core concepts through practical, application-oriented projects, prioritizing applied engineering and systems integration over purely abstract theoretical proofs. Students will begin by exploring the foundations of robotic spatial description and homogeneous transformations to understand how robots map and move in 3D space, applying forward kinematics to calculate precise robotic positioning, alongside a high-level overview of inverse kinematics, dynamics, and control. The curriculum then transitions to visual perception, covering image formation and classical computer vision techniques to effectively process raw camera data. To analyze this visual information, the course provides a review of neural network foundations before diving into convolutional neural networks (CNNs). Building on these deep learning principles, students will study object detection architectures to identify items within environments, object tracking to monitor their movement over time, and cutting-edge approaches like vision transformers and advanced segmentation models for detailed, pixel-level understanding. Finally, the course culminates in a survey of modern embodied AI, examining how high-level planning, vision-language models, and strict safety constraints are brought together to create reliable, fully autonomous systems.

Course Outline

- » **Foundations of robotic spatial description:** robot components; coordinate frames; pose (R,p); rotation matrices; Euler angles; quaternions.
- » **Homogeneous transformations:** SE(3); transforms; chaining and frame arithmetic.
- » **Kinematics fundamentals (with Dynamics & Control Overview):** joint space vs. task space; simplified DH convention; computing end-effector pose; high-level overview of inverse kinematics, dynamics, and control.
- » **Image Formation and Classical Computer Vision:** camera model; extrinsics; projection; calibration and distortion concepts, sampling; color spaces; convolution; filtering; gradients; edges; morphology.
- » **Neural network foundations (Review):** perceptron; MLP; activations; loss functions; backpropagation; optimization; regularization.
- » **Convolutional neural networks:** convolution/padding/stride; receptive fields; pooling; transfer learning.
- » **Object detection architectures:** localization vs. classification; IoU and mAP; two-stage vs. one-stage detection; YOLO-style inference.
- » **Object tracking and video foundations:** optical flow concepts; data association; classical and learning-based tracking.
- » **Transformers in vision (Overview):** self-attention; positional encodings; Vision Transformers; inductive bias.
- » **Advanced segmentation:** semantic vs. instance segmentation; Mask R-CNN overview; promptable segmentation and foundation models.
- » **Modern embodied AI (Survey):** vision-language models; high-level planning vs. low-level control; reliability and safety constraints.

Assessment

- ✓ Assignments and mini-projects: 8 points
- ✓ Exam(s): 8 points
- ✓ Final project: 4–8 point

References (Selected)

-  Spong, M. W., Hutchinson, S., & Vidyasagar, M. (2020). Robot modeling and control (2nd ed.). Wiley.
-  Taghirad, H. D. (2025). Fundamentals of robotics: Applied case studies with MATLAB® & Python (1st ed.). CRC Press.
-  Taghirad, H. D., & Khosravi, M. A. (2024). Introduction to robotics (1st ed.) [Book in Persian]. K. N. Toosi Uni. of Tech. Press.
-  Dellaert, F., & Hutchinson, S. (2023). Introduction to robotics and perception (Online book).
-  Corke, P. (2023). Robotics, vision and control: Fundamental algorithms in Python (3rd ed.). Springer.
-  Szeliski, R. (2022). Computer vision: Algorithms and applications (2nd ed.). Springer.
-  Torralba, A., Isola, P., & Freeman, W. T. (2024). Foundations of computer vision. The MIT Press.
-  Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep learning. The MIT Press.